

Gaurav Batra

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EDUCATION

University of Wisconsin-Madison (GPA: 3.92/4)

Masters of Science in Computer Science

Madison, WI, USA

Aug. 2024 – May 2026

IIIT Hyderabad (CGPA: 9.48/10)

B-Tech (Honours) in Computer Science And Engineering

Hyderabad, TN, IN

Aug. 2017 – Aug 2021

EXPERIENCE

Sigma Computing

Software Engineering Intern - AI

May 2025 – Aug 2025

San Francisco, CA, USA

- Built and launched a semantic search feature in Ask Sigma, allowing natural language queries to find a workbook, driving a **30%** improvement over Sigma's older search system.
- Deployed the feature as a scalable service on Kubernetes, optimizing for performance and reliability, resulting in **100+** daily queries served with **10%** lower latency.
- Partnered closely with the VP of AI and Co-founder, to deploy the feature in the production environment along with showcasing it in the **Sigma's 2025 AI product launch**.

Couture.ai

AI Platform Engineer

June 2021 - July 2024

Bengaluru, KA, India

- Core platform engineer responsible for designing, building, and scaling Couture.ai MLOps platform to manage the end-to-end machine learning lifecycle.
- Optimized the legacy pipelines by writing the latency-critical components in Rust, achieving **40%** reduction in deployment time and **20%** improvement in inference throughput.
- Spearheaded deployment of a secure, on-prem MLOps stack for the platform's first U.S. client on AWS EKS, driving enterprise compliance and directly contributing to a **\$250k ARR**.
- Engineered **15+** microservices in Python (FastAPI) and Golang, deployed with Kubernetes, and automated MLOps workflows with CI/CD across AWS and GCP, reducing manual deployment by **90%**.
- Designed core architectures, including the structuring of model registry, connector modules, and long-term roadmap for multi-tenant MLOps capabilities.
- Mentored two interns in developing a new ML observability stack using Grafana Loki; the resulting real-time log aggregation reduced critical production issue discovery time by **75%**.

Nvidia

Software Engineer Intern - AI

April 2020 - June 2020

Pune, MH, India

- Engineered a TensorFlow-based Convolutional Neural Network for optimized image compression, inspired by Better Compression with Deep Pre-Editing. Achieved a **15%** improvement in Peak Signal-to-Noise Ratio scores and a **10%** increase in SSIM metrics, outperforming traditional JPEG methods.
- Optimized video encoding by reducing bits-per-pixel to maintain stable frame rates under bandwidth constraints. Conducted extensive experiments and model tuning, leading to **27%** performance improvement.

Machine Learning Lab

Undergraduate Researcher

May 2019 - August 2021

Hyderabad, TN, India

- Authored the paper **Multiclass Classification under Dilute Bandit Feedback**, published at **PRICAI'21**. Showed that ML models trained with partial feedback can achieve accuracy comparable to full-feedback methods.

PROJECTS

Custom Filesystem using FUSE | C

Fall, 2024

- Designed and implemented WFS, a custom block-based filesystem using the FUSE framework, supporting basic filesystem operations and RAID 0/1 configurations to enhance fault tolerance and performance.

Deduplication Pipeline & Multi-GPU LLM Fine-Tuning | Python

Spring, 2025

- Developed a high-throughput deduplication pipeline using conventional hashing and MinHash to eliminate exact and near-duplicates from over **10M** documents across GitHub and Common Crawl sources.
- Built a multi-GPU fine-tuning pipeline with Hugging Face Accelerate and Transformers to efficiently fine-tune LLMs featuring optimized I/O, cluster-level parallelism, checkpointing, and reproducible W&B logging.

TECHNICAL SKILLS

Programming Languages: Python, C++, Golang, Rust, Bash, sed, AWK, SQL, Spark/PySpark

Software Skills: Fastapi, Flask, Docker, Kubernetes, PostgreSQL, $\text{\LaTeX}$, Redis, MATLAB, Celery, Istio Service Mesh

ML and DL Tools: Pytorch, Pandas, NumPy, Accelerate, Hugging-Face transformers, Weights and Biases.